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Supplementary information

Short- and medium-term impacts of strict anti-contagion policies on non-COVID-19 mortality in China

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Supplementary Materials for

Short- and Medium-Term Impacts of Strict Anti-Contagion Policies on Non-COVID-19 Mortality in China

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Supplementary Materials and Methods

Supplementary Materials and Methods 1

DSP data: The Disease Surveillance Points (DSPs) system is managed by the Chinese Center for Disease Control and Prevention. The system collects death records from the surveillance locations to understand death and disease patterns in China. The system was established in 1978 and has gradually increased its geographical coverage over the past four decades. In 2013, the system got a major upgrade and expanded its coverage from 161 points to 605 points, making the data representative at both the provincial and national levels. Each surveillance point represents a district (if in urban areas) or a county (if in rural areas). For each surveillance point, deaths that occurred both in hospitals and homes are reported, and the causes of death are determined according to a standard protocol by trained staff located in local hospitals or CDC branches. The DSPs system covers more than 324 million people in China, which accounts for 24.3% of the country's population.[20,21] The quality control procedures include annual training on standard workflow, random checking of the accuracy of disease classification and duplication, retrospective surveys on underreporting, and logic checks on the completeness and accuracy of disease codes. These quality checks are done at the county, province, and national levels.

In addition to the DSPs system, the Chinese CDC also manages the Communicable Disease Surveillance (CDS) system. The DSPs system is used for death registration, while the CDS system is used to monitor the development of infectious diseases, including COVID-19. These two systems are managed by different subsidiaries in the Chinese CDC and have separate administrative structures. Because the purpose of the DSPs system is to understand the overall

cause-of-death patterns in China, and because the death registration process carries no political stakes, there is no incentive for local hospitals or CDC branches to hide regular death information from the central government.

The SAP data: Most of the cities' strict anti-contagion policies (SAPs) were directly issued by the city-level governments, while a few were promulgated by the provincial governments. There are two types of SAPs: city SAPs and community SAPs. The former is defined as human mobility being restricted across different cities, and the latter is defined as mobility being restricted within a city. At the early stage of the outbreak, to prevent the virus from spreading outside Hubei province, the SAPs at the city level were adopted in Wuhan and its neighboring cities. The purpose of city SAPs was to restrict people in the epicenter of coronavirus from traveling to other cities. Later, as more cases were identified in other cities, the SAPs at the community level were implemented to further control the spread of the coronavirus within cities. The time lag between city SAPs and community SAPs was typically one to two weeks.

Supplementary Notes

Supplementary Note 1: Tests for pre-treatment parallel trends

The underlying assumption for the DiD model is that cities with SAPs and cities without SAPs would have parallel trends in mortality without the policies. To test how likely this assumption is to hold, we conduct an event study and investigate whether the two groups of cities have parallel pre-treatment mortality (Equation (3)). Fig. 3 plots our findings.

In Panel A, we compare the total number of non-COVID-19 deaths between the treatment and control groups before and after the SAPs implementation. The difference between the two groups three weeks before the SAPs is set as the reference level (*i.e.*, the zero coefficient for week -3), so the post-SAP effects are relative to the period three weeks before the launch of the policy. For most outcome variables, we do not observe systematic pre-trends between the two groups before the city/community SAPs. This finding implies that the parallel trend assumption is likely to hold in our setting. In comparison, the trends break after the SAPs were enforced, *i.e.*, the lagged terms ($k \geq 1$) become negative and statistically significant. In addition, we observe that the difference becomes larger as more lags are included, suggesting an accumulating health benefit of city/community SAPs.

In Panels B-F of Fig. 3, we repeat this exercise to investigate the trends in deaths from cardiovascular diseases, injuries, ALRI, CLRI, and neoplasms before and after the SAPs implementation. The results suggest that the parallel trend assumption holds during the pre-treatment period for all these outcomes as well. Note that deaths from injury (Fig. 3C) and ALRI (Fig. 3D) start to decline a week prior to the policy implementation. We think this is because people had started to change their behaviors about one week before the SAPs were formally implemented.

Supplementary Note 2: Robustness checks

We conduct a variety of robustness checks and show that our results are not qualitatively affected by several decisions we make in the baseline analysis (Extended Data Fig. 4). First, we add

weather and socio-economic controls into the baseline regressions (Row 2). Weather variables include daily average temperature, humidity, wind speed, and air pressure. Socio-economic status includes time-invariant per capita GDP, number of hospital beds per thousand people, and share of employment in secondary industries. These variables interact with a third-order polynomial function of time in the regressions. The results are quantitatively unchanged.

Second, we examine whether the baseline characteristics of the treatment DSPs (with SAPs) and control DSPs (without SAPs) are different. In Supplementary Table 3, we find that cities with a larger number of COVID-19 cases were more likely to enforce the SAPs, while other factors (non-COVID-19 mortality and socio-economic status before the pandemic) are not correlated with the decision on SAP implementation. In addition, we adopt propensity score matching based on these variables before running the DiD model and find that results remain largely similar (Extended Data Fig. 4 Row 3).

Third, we weight regressions by population in each DSP (Row 4). Intuitively, this allows us to estimate the SAPs' effects on an average individual instead of an average DSP. Without weighting, cities with smaller populations could drive the baseline results. We find that the results remain very similar.

Fourth, we exclude 22 DSPs in Hubei province from the regression (Row 5). COVID-19 was first identified in Wuhan city in Hubei province. Thus, the DSPs in the province could be very different. We find that the results are quantitatively similar to the baseline after excluding data from Hubei.

Fifth, we also include the three DSPs (districts) located in Wuhan in the regressions and find the effects become slightly smaller (Row 6). In addition, the sign for non-COVID-19 acute lower

respiratory infections (ALRI) is reversed and becomes positive (statistically insignificant) (Row 6 of Column D). This finding is consistent with a recent paper suggesting that some COVID-19 deaths in Wuhan could have been misclassified as ALRI or pneumonia.[15] The results also help justify our decision to exclude Wuhan from the baseline analyses.

Six, we use different definitions of SAPs (Row 7). Shortly after the COVID-19 outbreak, a dozen cities around Wuhan (covering 65 DSPs) restricted human mobility across cities (city SAPs), mostly in late January. Later, as the virus started to spread outside Hubei province, more cities began to restrict mobility within a city (community SAPs). In our baseline analysis, the SAP status is equal to one if either city SAPs or community SAPs were implemented. Alternatively, we can use whether the community SAPs were implemented to define the SAP status. The results show that the mobility restriction within cities reduces deaths from cardiovascular disease (CVD), acute lower respiratory infection (ALRI), and chronic lower respiratory infection (CLRI), while it has small effects on deaths from injury.

Finally, we use natural logarithms of the number of deaths as outcome variables and weight regressions using populations; our findings are qualitatively similar, while the magnitude becomes slightly smaller (Row 8).

Supplementary Tables

Supplementary Table 1. Summary statistics: January 1 to April 7

Panel A: Daily Mortality (Person)			
	Total	9.62	[7.48]
	Cardiovascular disease (CVD)	4.75	[4.13]
	Injury	0.56	[0.93]
	Acute lower respiratory infection (ALRI)	0.13	[0.42]
	Chronic lower respiratory infection (CLRI)	0.74	[1.18]
	Neoplasms	2.17	[2.27]
	Others	1.20	[1.43]
Panel B: Other Variables			
	GDP per capita (10,000 RMB)	5.97	[3.89]
	Number of hospital bed per thousand people	15.95	[31.33]
	Share of employment in secondary industry (%)	24.20	[12.00]
	Air Quality Index	69.98	[46.50]
	PM2.5 Concentration ($\mu g/m^3$)	46.91	[37.78]
	PM10 Concentration ($\mu g/m^3$)	69.25	[60.62]
	# of DSP counties	602	

Notes: This table reports the means and standard deviations of the key variables. All DSP districts/counties are included in the analysis except three from Wuhan.

Supplementary Table 2. Event study estimation results

	(1)	(2)	(3)	(4)	(5)	(6)
	Total # of Deaths	CVD	Injury	Acute Lower Respiratory Infection	Chronic Lower Respiratory Infection	Neoplasms
>= 5 Weeks Before	-0.102	-0.050	-0.013	-0.012	-0.085**	0.036
	[0.173]	[0.111]	[0.028]	[0.015]	[0.039]	[0.055]
4 Weeks Before	-0.123	-0.055	-0.032	0.004	-0.022	0.000
	[0.113]	[0.072]	[0.023]	[0.013]	[0.027]	[0.044]
2 Weeks Before	-0.064	0.007	-0.015	-0.009	-0.007	-0.038
	[0.114]	[0.076]	[0.024]	[0.012]	[0.028]	[0.042]
1 Week Before	0.063	0.082	-0.059**	-0.030**	-0.005	0.044
	[0.137]	[0.091]	[0.024]	[0.013]	[0.034]	[0.050]
Week of Implementation	-0.048	-0.003	-0.063**	-0.029**	0.014	0.019
	[0.152]	[0.094]	[0.027]	[0.015]	[0.036]	[0.052]
1 Week Later	-0.157	-0.117	-0.073***	-0.022	-0.007	0.038
	[0.162]	[0.101]	[0.027]	[0.015]	[0.034]	[0.052]
2 Weeks Later	-0.369**	-0.217**	-0.086***	-0.022	-0.067*	0.034
	[0.165]	[0.102]	[0.028]	[0.015]	[0.036]	[0.055]
3 Weeks Later	-0.615***	-0.381***	-0.041	-0.044***	-0.084**	-0.058
	[0.181]	[0.114]	[0.028]	[0.015]	[0.040]	[0.058]
4 Weeks Later	-0.659***	-0.385***	-0.069**	-0.040***	-0.100***	-0.062
	[0.162]	[0.107]	[0.028]	[0.015]	[0.038]	[0.052]
5 Weeks Later	-0.990***	-0.536***	-0.085***	-0.055***	-0.139***	-0.099*
	[0.183]	[0.115]	[0.029]	[0.014]	[0.041]	[0.053]
>= 6 Weeks Later	-0.982***	-0.559***	-0.060**	-0.052***	-0.159***	-0.079
	[0.197]	[0.122]	[0.027]	[0.013]	[0.044]	[0.053]
DSP Fixed Effect	YES	YES	YES	YES	YES	YES
Date Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	58,996	58,996	58,996	58,996	58,996	58,996
Adjusted R-Square	0.763	0.654	0.322	0.202	0.433	0.544
# of DSP Counties	602	602	602	602	602	602

Notes: Each column represents a separate regression using the event-study approach (Equation (3)). The dependent variable is the number of deaths from a specific cause. We include leads and lags of the start of SAPs in the regressions to test the parallel trend assumption. The dummy variable indicating three weeks before the SAPs was omitted from the regressions. We use mortality data from January 1 to April 7, 2020. All DSP districts/counties are included in the analysis except three from Wuhan. The standard errors clustered at the prefectural level are reported below the estimates. *P < 0.1, **P < 0.05, *** P < 0.01.

Supplementary Table 3: Examination of endogeneity with implementation of SAPs

Dependent variable: Implementation of SAPs	(1)	(2)
Log (COVID-19 cases)	0.06***	0.06***
	[0.01]	[0.01]
Mortality rate in 2019	-0.12	-0.10
	[0.30]	[0.31]
Mortality rate from CVD in 2019	0.09	0.06
	[0.32]	[0.33]
Mortality rate from injuries in 2019	-0.42	-0.49
	[0.43]	[0.46]
Mortality rate from Acute Lower Respiratory Infection in 2019	0.22	0.31
	[0.46]	[0.48]
Mortality rate from Chronic Lower Respiratory Infection in 2019	-0.01	-0.05
	[0.33]	[0.34]
Mortality rate from neoplasms in 2019	0.22	0.21
	[0.36]	[0.37]
Mortality rate from other diseases in 2019	-0.26	-0.26
	[0.53]	[0.53]
GDP per capita		-0.23
		[0.39]
Number of hospital beds per 1,000		-0.38
		[0.47]
Secondary industry share in employment		0.17
		[0.18]
Air Quality Index in 2019		0.00
		[0.00]
Obs.	602	602
Province Fixed Effect	YES	YES
Adjusted R-Square	0.571	0.574
# of DSP Counties	602	602

Notes: All DSP counties are included in the analysis except the three in Wuhan. The outcome variable is a dummy indicating whether a DSP's city had implemented SAPs. Log (COVID-19 cases) denotes logarithms of the number (plus 1) of confirmed COVID-19 cases on February 1 2020. Since COVID-19 data are available only after January 23 2020, we use data on February 1, when less than 10% of cities had implemented SAPs. All regressions include province dummies. *P < 0.1, **P < 0.05, *** P < 0.01. Robust standard errors are used.

Supplementary Table 4: The short- and long-term impacts of SAPs on deaths from different causes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total # of Deaths	CVD	Injury	Acute Lower Respiratory Infection	Chronic Lower Respiratory Infection	Neoplasms	Other Causes
	Mean = 8.625	Mean = 4.131	Mean = 0.561	Mean = 0.111	Mean = 0.605	Mean = 2.098	Mean = 1.058
SAPs Implemented	-0.448***	-0.277***	-0.035**	-0.020**	-0.037	-0.055*	-0.023
	[0.124]	[0.075]	[0.014]	[0.009]	[0.025]	[0.032]	[0.025]
Post-SAPs	-1.081***	-0.628***	-0.047**	-0.044***	-0.107***	-0.125**	-0.119***
	[0.221]	[0.132]	[0.019]	[0.010]	[0.040]	[0.051]	[0.035]
DSP Fixed Effect	YES	YES	YES	YES	YES	YES	YES
Date Fixed Effect	YES	YES	YES	YES	YES	YES	YES
Obs.	128,226	128,226	128,226	128,226	128,226	128,226	128,226
Adjusted R-Square	0.735	0.620	0.299	0.181	0.381	0.537	0.355
# of DSP Counties	602	602	602	602	602	602	602

Notes: Each cell in the table represents a separate DiD regression. All DSP districts/counties are included in the analysis except three from Wuhan. The outcome variable is the daily number of non-COVID-19 deaths from the DSP districts/counties. We use mortality data from January 1 and July 31, 2020 for this analysis. The explanatory variable for short-term SAPs is a dummy indicating whether a DSP's city had SAPs in place on a particular date between January 24 to April 7, 2020. The medium-term explanatory variable is an interaction between a dummy indicating whether the DSP had ever had SAPs in place and the period between April 8 and July 31 2020. The standard errors clustered at the DSP level are reported below the estimates. *P < 0.1, **P < 0.05, *** P < 0.01.

Supplementary Table 5: The heterogeneous impacts of SAPs on deaths

	(1) Total	(2) Total	(3) Total	(4) Total	(5) Total	(6) Total
SAPs	-0.410***	-0.267***	-0.033**	-0.021**	-0.044*	-0.034
	[0.107]	[0.066]	[0.014]	[0.008]	[0.024]	[0.029]
SAPs*GDP per capita	-0.155	-0.028	-0.008	-0.018**	-0.028	-0.033
	[0.108]	[0.057]	[0.010]	[0.008]	[0.024]	[0.025]
SAPs*Hospital beds per capita	0.004	0.040	-0.015*	-0.008	-0.005	-0.005
	[0.084]	[0.049]	[0.008]	[0.008]	[0.016]	[0.021]
SAPs*Share of employment in secondary industry	-0.120	-0.086*	-0.011	0.004	-0.026	0.026
	[0.085]	[0.049]	[0.010]	[0.006]	[0.017]	[0.022]
SAPs*Air Quality Index	-0.401***	-0.271***	0.000	-0.016***	-0.086***	-0.025
	[0.104]	[0.059]	[0.010]	[0.005]	[0.019]	[0.023]
SAPs*Base mortality rate	-0.336***					
	[0.109]					
SAPs*Base mortality rate from CVD		-0.153**				
		[0.067]				
SAPs*Base mortality rate from injury			-0.039***			
			[0.010]			
SAPs*Base mortality rate from Acute Lower Respiratory Infection				-0.058***		
				[0.011]		
SAPs*Base mortality rate from Chronic Lower Respiratory Infection					-0.146***	
					[0.023]	
SAPs*Base mortality rate from Pneumonia						-0.022
						[0.033]
DSP Fixed Effect	YES	YES	YES	YES	YES	YES
Date Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	58,996	58,996	58,996	58,996	58,996	58,996
Adjusted R-Square	0.764	0.655	0.322	0.206	0.436	0.544
# of DSP Counties	602	602	602	602	602	602

Notes: Each cell in the table represents a separate DiD regression. We interact the treatment dummy with baseline socio-economic variables to understand the heterogeneity impacts of SAPs. The socio-economic variables are standardized to have a mean of 0 and a standard deviation of 1. We use mortality data from January 1 to April 7, 2020. All DSP districts/counties are included in the analysis except three from Wuhan. The standard errors clustered at the DSP level are reported below the estimates. *P < 0.1, **P < 0.05, *** P < 0.01.

Supplementary Table 6: The implementation of SAPs in Chinese cities

City Name	City SAPs	Community SAPs	City Name	City SAPs	Community SAPs
Huanggang	2020/1/23	2020/2/1	Wuhai	2020/2/12	2020/1/28
Ezhou	2020/1/23	2020/2/4	Lvliang		2020/1/29
Wuhan	2020/1/23	2020/2/10	Ganzhou	2020/2/6	2020/1/30
Xianning	2020/1/24	2020/2/5	Wuzhong		2020/1/31
Yichang	2020/1/24	2020/2/10	Sanmenxia		2020/1/31
Municipal County	2020/1/24	2020/2/10	Yinchuan	2020/1/31	2020/1/31
Jingmen	2020/1/24	2020/2/10	Xinyang	2020/2/6	2020/1/31
Shiyan	2020/1/24	2020/2/10	Lishui		2020/2/1
Enshi	2020/1/24	2020/2/10	Anshun	2020/2/5	2020/2/1
Jingzhou	2020/1/24	2020/2/10	Southeast Guizhou		2020/2/2
Xiaogan	2020/1/24	2020/2/10	Xinzhou		2020/2/2
Huangshi	2020/1/24	2020/2/11	Jinhua		2020/2/2
Qinhuangdao	2020/1/25	2020/2/8	Liupanshui		2020/2/2
Tangshan	2020/1/28	2020/2/6	Guiyang		2020/2/2
Xiangyang	2020/1/28	2020/2/10	Yulin		2020/2/2
Dongying	2020/1/30	2020/2/10	Wenzhou	2020/2/2	2020/2/2
Chongqing	2020/1/31	2020/2/8	Fangchenggang	2020/2/8	2020/2/2
Jining	2020/2/3	2020/2/7	Bayannur	2020/2/12	2020/2/2
Ningbo	2020/2/4	2020/2/5	Hohhot	2020/2/12	2020/2/2
Zaozhuang	2020/2/4	2020/2/12	Xilin Gol League	2020/2/12	2020/2/2
Panjin	2020/2/5	2020/2/6	Zunyi		2020/2/3
Fuxin	2020/2/5	2020/2/6	Guigang		2020/2/3
Dalian	2020/2/5	2020/2/6	Jincheng		2020/2/3
Fushun	2020/2/5	2020/2/6	Huai'an		2020/2/3
Chaoyang	2020/2/5	2020/2/6	Binzhou		2020/2/3
Jinzhou	2020/2/5	2020/2/6	Taizhou		2020/2/3
Tieling	2020/2/5	2020/2/6	Southwest Guizhou		2020/2/3
Shenyang	2020/2/5	2020/2/6	Zhoushan		2020/2/3
Yangzhou	2020/2/5	2020/2/6	Fuzhou	2020/2/6	2020/2/3
Dandong	2020/2/5	2020/2/6	Wuxi	2020/2/9	2020/2/3
Liaoyang	2020/2/5	2020/2/6	Ulanqab	2020/2/12	2020/2/3
Municipal District	2020/2/9	2020/2/10	Erdos	2020/2/12	2020/2/3
Shuangyashan		2020/2/4	Lu'an		2020/2/5
Zigong		2020/2/4	Maoming		2020/2/5
Zhenjiang		2020/2/4	Huaihua		2020/2/5
Anshan		2020/2/4	Ganzi		2020/2/5
Lianyungang		2020/2/4	Liuzhou		2020/2/5
Wuhu		2020/2/4	Suqian		2020/2/5
Songyuan		2020/2/4	Kaifeng		2020/2/5
Huainan		2020/2/4	Meizhou		2020/2/5
South Guizhou		2020/2/4	Pingdingshan		2020/2/5
Tongren		2020/2/4	Nanchong		2020/2/5
Bengbu		2020/2/4	Quzhou		2020/2/5
Nanyang		2020/2/4	Lijiang		2020/2/5
Xi'an		2020/2/4	Heihe		2020/2/5

Supplementary Table 6: The implementation of SAPs in Chinese cities

City Name	City SAPs	Community SAPs	City Name	City SAPs	Community SAPs
Wenshan		2020/2/4	Suizhou		2020/2/5
Dezhou		2020/2/4	Fuyang		2020/2/5
Fuzhou	2020/2/4	2020/2/4	Weifang		2020/2/5
Hangzhou	2020/2/4	2020/2/4	Huzhou		2020/2/5
Harbin	2020/2/4	2020/2/4	Zhuhai		2020/2/5
Zhumadian	2020/2/4	2020/2/4	Hengshui		2020/2/5
Nantong	2020/2/4	2020/2/4	Guilin		2020/2/5
Changzhou	2020/2/4	2020/2/4	Meishan		2020/2/5
Zhengzhou	2020/2/4	2020/2/4	Daxinganling		2020/2/5
Linyi	2020/2/4	2020/2/4	Yichun		2020/2/5
Jingdezhen	2020/2/4	2020/2/4	Quanzhou		2020/2/5
Nanjing	2020/2/4	2020/2/4	Liaocheng		2020/2/5
Xuzhou	2020/2/4	2020/2/4	Zhoukou		2020/2/5
Jiujiang	2020/2/6	2020/2/4	Daqing		2020/2/5
Yingtan	2020/2/6	2020/2/4	Jiaxing		2020/2/5
Huaibei	2020/2/9	2020/2/4	Yancheng		2020/2/5
Bijie		2020/2/5	Sanya		2020/2/5
Hechi		2020/2/5	Zhaoqing		2020/2/5
Haikou		2020/2/5	Luzhou		2020/2/5
Wuzhou		2020/2/5	Chengde		2020/2/5

Notes: Panel A lists the cities that were under city SAPs earlier than community SAPs. Panel B lists the cities that had community SAPs earlier than city SAPs. The effective dates of the two SAPs are both listed.